

-
-
-
-
-

Chapter 8

Warehousing Overview



Introduction

Warehousing provides different levels of support for different organisations, or different warehouses within the same organisation. You can use Warehousing to provide a tracking system for a limited range of key items. You can expand this to encompass all aspects of warehouse control and management.

The map of the warehouse can include proximity algorithms. You use these algorithms as one way of deciding where you locate items.

You can base picking and marshalling by order, journey, customer, and due date. Warehousing takes account of expiry dates, lots or batches, pallet numbers and warehouse organisation when allocating stock to picking instructions.

Warehousing is designed to fit into the operational environment of a warehouse, you only need to enter the minimum of data to operate it. All you normally need to do is confirm the lists of tasks. The database easily interfaces to technical advances in warehousing such as hand-held terminals.

Relationships to Other Modules

The only module which is a pre-requisite for the operation of Warehousing is Inventory Management.

Warehousing is designed to be used in conjunction with, and enhance the following modules:

- Sales Order Processing
- Purchase Management
- Production Definition Management
- Production Order Control
- Production Schedule Control.

Once you have installed Warehousing and fully activated a stockroom as a warehouse, links are switched on between the operational modules and the warehouse to provide additional location management and control.

You can automatically print pallet labels for items received against a purchase order. The labels can include put-away instructions. You record despatches for sales orders in Sales Order Processing and you can pick, marshall, and despatch, issues for production orders and schedules. You can make receipts from production orders and schedules into a warehouse and start put-away routines.

Warehousing Configuration

You can operate Warehousing for a number of companies, keeping the characteristics of each on control files. Warehousing uses the control data stored for Inventory Management. The precise control, however, is at warehouse level because you could have different policies in different parts of the organisation.

The policies you control at each warehouse are:

- How you give instruction and record actions.
- Check digits used to confirm actions.
- Stock rotation.
- The sequence in which to produce action lists.

All actions resulting in an Inventory receipt automatically locate the item in the receiving location. Similarly, when you require picking, Warehousing recommends that you move stock for items from certain locations to the marshalling location before issue.

You can define any locations set up as the marshalling and receiving location. However, we recommend that you set up warehouse areas so that each contains a single location with a Location Code which is the normal name for receiving and marshalling in the warehouse. You can set up these areas to satisfy the requirements of each individual warehouse.

Accuracy and discipline are the fundamentals of the warehousing operation. If warehousing detects an error, it directs a message instantly to a nominated user.

You must set up for each item managed by Warehousing an item/warehouse profile that defines its rules and characteristics. You can set up defaults for the warehouse to minimise the amount of data entry required when you maintain these profiles.

Since a user of Warehousing is normally at a particular site, you can define a default warehouse. This makes the day-to-day running easier particularly for recording transactions and making enquiries.

Maintenance of Reference Data

You set up much of the data for Warehousing when you install the base modules. You must, however, enter additional information to get the maximum benefit from Warehousing.

You can maintain the following information:

Warehouse Profiles

Use the warehouse profile to:

- Set up the warehouse policies, such as confirmation check digits.
- Set up the special areas for receiving and marshalling.
- Define who receives the error messages.
- Define defaults used in the item/warehouse profiles.
- Activate or de-activate the warehouse.

Area Profiles And Dimensions

These define the shape and structure of the warehouse. You define a series of areas, each of which is divided into a number of locations. For example, you could define three areas, Floor used for the receiving and marshalling locations, Main Store to represent most of the aisles, rows and bays in the warehouse and Picking to represent the pick locations.

You use up to four dimensions to define the area, such as aisle, bay, row. The dimensions are used to build up the code that identifies individual locations. Each code can be up to six characters in length, but the final location code (the concatenation of all the dimension codes) cannot exceed twelve characters.

Item/Warehouse Profiles

You must associate each item kept in a warehouse with a location. Warehousing makes recommendations for where to put items, when and from where replenishment should take place, and from where items should be picked. You must define a profile for each item in each warehouse. The profile contains five type of information:

- Packaging - This relates the item in inventory stockroom issue units of measure to the packaging types used in the warehouse. It also defines the basis of stock rotation.
- Fixed Locations.
- Put-Away - This defines the steps Warehousing uses when making recommendations as to where an item should be put.
- Commit - This defines the steps Warehousing uses when making recommendations of where an item should be picked from.
- Pick Location - You can set up information including replenishment trigger and the replenishment quantity. You can also define the maximum, in a single pick, to take from this location before the pick is made direct from bulk.

Packaging Types

You move many items in a warehouse using pallets or containers of some kind. You must define these to Warehousing and then create the relationship between the pack and item. You must set up a default pack type for each item.

Location Types

You chose the size, shape, and use of locations to meet particular storage requirements, including the number and distribution of pack types. Warehousing uses this, together with the number of packs per location, to decide whether to put away a receipt in a location. You must then define the distribution of location types throughout the warehouse areas. Also you can remove locations from use by tagging them as void, and limit their use to single-item.

Warehouse Map

The warehouse map assigns location types to locations to determine the characteristics of the locations. For example, the locations within the Main Store area can have a location type of Standard Pallet Racking which limits the locations to storing only one pallet at a time.

Action Rules

You should only move an item in a warehouse as a result of an instruction. You must decide the best location for the item depending upon factors such as:

1. The item itself.
2. How it is packed.
3. Does it require special conditions?
4. Is the item picked from a picking location?
5. Can mixed lots be put in the same location?
6. Must the receiving area be cleared quickly?

You can create Warehousing rules to mimic these decisions. You can set up the following to control the recommendations for put-away, pick and replenish:

1. Use the pick location.
2. Use the fixed locations.
3. Use an item rule. This reproduces a line-of-sight check. To create the rule you must define a series of locations to test in relation to the pick location.
4. Use an area rule. This type of rule can make use of the proximity data set up for an area or the randomising algorithm. This rule reviews a large number of locations to answer such questions as:
 - Find the first not-full location in this part of the warehouse, closest to the receiving bay
 - Select randomly a not-full location.
5. Use stock rotation dates, such as expiry and arrival date.

Resources

Resources are warehouse equipment, such as trolleys and fork lift trucks. You can define a resource then relate it to pack types within a location type. For example, you can relate trolley to a pack type of pallet in a location type of floor.

Check Digits

You use check digits or return codes to make sure that the warehouse staff have visited the defined location. You confirm the action by entering the check digit. For an intra-warehouse movement, you must enter both check digits. To use check digits:

1. You set up the warehouse to use check digits.
2. Set up a set of default check digits for the warehouse. You can override check digits for each or all areas.
3. If you use check digits, but do not set up a default, Warehousing uses modulus 23 to generate digits.

Processing

You can divide most of the work in a warehouse into three types of tasks:

- Putting away items.
- Moving items to optimise storage, picking and despatching.
- Picking and despatching.

Warehousing makes recommendations for each of the three tasks, and produces instruction lists on paper and on screen. You can confirm these tasks have been completed by entering 1 against the action, or, if you use check digits, entering the appropriate digits. If you do not use check digits, you can set Warehousing to confirm actions by default. You can override those actions.

You can change actions to show you used different locations when carrying out the instructions.

You can carry out some of the processing in a background subsystem or in batches. Running jobs in batches can be especially useful for printing, where special stationery for work planning is required.

Receiving and Putting Away

When you receive and put away an item you must record the specific actions and process the recommendations made by Warehousing.

The steps to receive an item and put it away are:

Inventory Receipt

Record the inventory receipt in one of the operational modules, Purchase Management, Inventory Management, Production or Sales Order Processing. This increases the inventory balance at the stockroom. You must therefore increase stock in the warehouse. The sub-system moves the item into the Receiving location. This balances the warehouse and inventory without an explicit additional action in the operational module. However, as Warehousing can operate with multiple packaging types, you can confirm or change the packaging type when you receive the item into a warehouse.

The receiving sub-system:

- Moves the item into the receiving location.
- Determines the number of packs.
- Identifies the rules for putting the item away.
- Makes enough recommendations to put the item away into the warehouse.
- Produces one recommendation for each unique combination of item, lot, packaging type and location.

Warehousing plans the item out of receiving and into the locations.

Printing Documents

Before you confirm an action, Warehousing must generate an instruction. If you requested pallet labels, Warehousing prints them, otherwise it a list. The labels and list only print in the warehouse when requested.

Confirmation of the Action

You can confirm put-away by entering a single character or check digit found at the location. You can use the sub-system to update of the database so the update is normally complete within seconds of finishing the entry.

Intra Warehouse Move

You must be able to record that you need to move stock within a warehouse and that you have moved stock. You may need to move stock in the warehouse because you are moving bulk to a picking area or refurbishing part of the warehouse. If the need for action is immediate, you can report the move to Warehousing after the event.

Warehousing produces a list of instructions which you can see on a screen and confirm or change.

Warehousing distinguishes between movements generated for replenishment and those manually requested. You can produce these on separate lists so you can assign the proper priority for the different actions. When you confirm the actions there are different overrides are appropriate for the different moves. For example, if this is a replenishment move, you cannot change the item or the move to location.

Picking and Despatching

You initially allocate stock to sales orders in Sales Order Processing or Production Control and Costing. This allocate against the available stock figure in the nominated stockroom but does not identify where to find the stock. You use Warehousing to find the stock.

Warehousing decides the stock location by reviewing such factors as:

- The quantity to despatch.
- Stock rotation.
- Physical proximity of location.
- The practicality of picking a number of order lines together.

The picking and despatching cycle has the following events:

Request Picking Lists

Warehousing considers requests to despatch generated from Sales Order Processing or requests to issue generated from Production Control and Costing. Warehousing performs a series of tasks to print of the documents necessary to pick and despatch the items. These are:

- Select a range or group of orders.
- Sequence and combine them.
- Decide from which locations to pick the stock.
- Print a pick list organised to suit the warehouse
- Print appropriate marshalling lists which cross-reference picking lists and despatch or issue notes.
- Print the despatch or issue notes.



Note: Kit parent and non-stock items are not committed and do not print on the warehouse pick list.

Confirm Pick

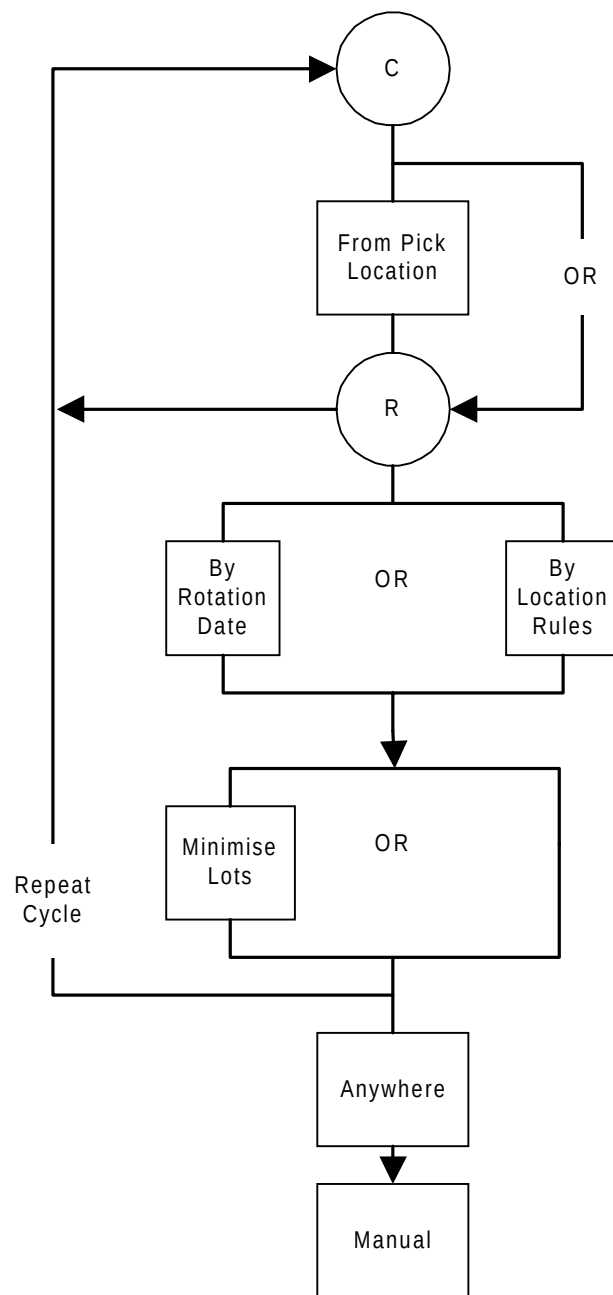
You can confirm the pick by entering either the check digit or return code. You can override the module instruction by changing both the location and the quantity. This will result in Warehousing recording an error. When you confirm pick, Warehousing transfers the stock to the marshalling location.

Confirm Despatch / Issue

You can confirm despatch from the marshalling location at order level using the standard picking note, despatch note or shop documentation produced by Sales Order Processing and Production control and costing.

You can produce reports to help marshall complex loads. The reports link picking lists and despatch or issue notes. If the despatch has taken place as intended, you can use a single function to confirm it. For despatches you can add delivery and freight charges, and reprint the despatch note with actual despatch quantities.

Commit and Replenish Cycle



Warehousing performs this process for each commit action. A commit action is at the level of item, lot, stock rotation date, pack type or pack number.

Enquiries

You can group enquiries into one of three categories:

Status

You can use status enquiries to display what is in a location, or where you store a location. You can also sequence the enquiries by location, area or item rule.

You can display:

- Item details
- Location contents
- Location details.

Action Lists

There are three types of action lists:

- Put away
- Move
- Pick.

You can display the unfinished lists. You can also display details of despatch instructions sent but not yet actioned and then access to the full order enquiry within Sales Order Processing.

Movements

You can display the warehouse movements and the reported error conditions. If an action is requested but not then confirmed as instructed, an error record is written. You can use multiple selection criteria for both movements and errors enquiries.

Reports

You can use selection and sequencing parameters to define a report before running a report. Standard reports print on standard listing paper with 132 print positions and 6 lines to the inch.

All reports print the procedure or program number in the top left, and the page number, date and time to the top right.

Implementation

Before implementing Warehousing you must consider:

- How Warehousing and other Modules should be used.
- How many warehouses should be set up.

Procedure

There are two possible implementation procedures:

- Warehousing with Inventory already installed
- Warehousing with Inventory newly installed

Use the appropriate procedure as a guide.

We recommend that whichever procedure you use, you follow a project plan. This is so you can capture and record the necessary data before entering the stock count.

The basic steps for a successful implementation are:

- Master file data collection and input.
- Work load organisation.
- Suspend processing of stock.
- Initial date take-on and reconciliation.
- Resume processing of stock.

Warehousing with Inventory Already Installed

This assumes that you are familiar with Inventory Management and you have been trained to use Warehousing.

Master File Data

Define the data required to make Warehousing functional. The amount of effort required depends upon the number of items, locations, pack and location type details which you need to define.

Work Load Organisation

It is essential that during the stock count exercise and the early stages of processing with Warehousing that you control the movement of stock. Try to minimise the activities for receiving and issuing stock.

Receipts

Review any receipts which have been booked into stock but have not been put away. Then:

- Put the items away, reducing the items at the receiving area to those not yet booked into stock.
- If the receipts were made through Purchase Management transfer them back to either goods inwards or inspection. Then receive the items into stock after activating the warehouse.
- Leave the items at the receiving location. Record the stock at Data Take-on and then move manually after activating the Warehouse.

Issues

Process all requests for issuing stock. Pick Items to despatch and move them to the Marshalling location. Where possible despatch the items and confirm the actions. If you leave items at Marshalling, you must enter the full details required for Confirming Despatch.

Suspend Processing of Stock

Suspend stock movement into, around and out of the warehouse whilst you perform and reconcile the stock count for Data Take-on. Complete any outstanding transactions.



Note: From this point until you activate the warehouse, suspend the processing of Inventory, Purchasing, Production and Sales Order Processing.

Initial Data Take On/Reconciliation

Do a full stock count including every item and every location in the Warehouse. Enter the details using Data Take-On programs.

Reconcile the count to the Inventory lot and stockroom balances, discrepancies are shown on the report. Verify any variances. Once the count is cleared as correct,

update the location details using the Reconciliation with Update option. This also makes appropriate adjustments to the Inventory balances.

Resume Processing of Stock

Activate the warehouse.

Resume transaction processing in all areas of the warehouse. You can record transactions in the modules previously suspended.

Receipts and issues are now subject to the control of Warehousing.

Warehousing with Inventory Newly Installed

You must undertake training so you can understand both Inventory Management and Warehousing.

Master File Data Collection and Input

Enter the data required to make both Inventory and Warehousing functional. Make sure that you enter dependencies in the appropriate order. The amount of effort required depends upon the number of items, location, pack and location type details that you need to define.

Work Load Organisation

During the stock count exercise and the early stages of processing with Warehousing, you control the movement of stock.

Review items that are in the receiving area. Where possible do the following:

- Put the items away before the stock count.
- Record, at data take-on, the items in the receiving area. You must enter manual moves after activating the warehouse.
- Leave the items at the receiving location, do not record the stock during data take-on. You receive the items after activating the warehouse.

Process items to despatch through to the marshalling area. It is better to start with as few incomplete despatches as possible.

Suspend Processing of Stock

Suspend the movement of stock whilst performing the stock count for Data Take-on.

Movement of stock can resume after activating the warehouse.

Initial Data Take On / Reconciliation

Do a full stock count of every item and location in the warehouse. Enter the details either through the Data Take-On programs or by recording Physical Stock Adjustments in Inventory.

If you use Data Take-On, use Reconciliation with Update to set the Inventory balances.

Resume Processing of Stock

Activate the warehouse.

You can now resume transaction processing in the warehouse.

Receipts and issues are now subject to the control of Warehousing.